Consolidated Teaching and Examination Scheme

B.Tech Chemical Engineering – 2024-28

MS: Mid Semester | IA: Internal Assessment | ES: End Semester | LW: Laboratory Work | LE/Viva: Laboratory Examination/Viva | CE: Continuous Evaluation | PP/NP: Passed/Not Passed

Semester 1													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA101T	Mathematics - I	3	1	0	4	4	25	25	50	--	--	100
24HS101T	English Communication	2	0	0	2	2	25	25	50	--	--	100
24CP101T	Computer Programming - I	1	0	0	1	1	25	25	50	--	--	100
24CP101P	Computer Programming - I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH101T	Engineering Chemistry	3	0	0	3	3	25	25	50	--	--	100
24CH101P	Engineering Chemistry Laboratory	0	0	2	2	1	--	--	--	50	50	100
24EE101T	Elements of Electrical and Electronics Engineering	3	0	0	3	3	25	25	50	--	--	100
24EE101P	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	2	1	--	--	--	50	50	100
24BT101T	Biological Systems for Engineers	2	0	0	2	2	25	25	50	--	--	100
24HS104T	Organizational Behaviour	1	0	0	1	1	25	25	50	--	--	100
24YOG101 / 24NSS101 / 24NCC101	Yog, Health & Hygiene / National Service Scheme (NSS) / National Cadet Corps (NCC)	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**1**	**8**	**24**	**20**						

Semester 2													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24HS105T	Professional Communication	2	0	0	2	2	25	25	50	--	--	100
24MA103T	Mathematics - II	3	1	0	4	4	25	25	50	--	--	100
24CP102T	Computer Programming - II	1	0	0	1	1	25	25	50	--	--	100
24CP102P	Computer Programming - II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PH103T	Modern Physics	3	0	0	3	3	25	25	50	--	--	100
24PH103P	Modern Physics Laboratory	0	0	2	2	1	--	--	--	50	50	100
24ME101P	Workshop Practices	0	0	2	2	1	--	--	--	50	50	100
24ME102P	Engineering Graphics	0	0	4	4	2	--	--	--	50	50	100
24CV101T	Environmental Science	2	0	0	2	2	25	25	50	--	--	100
24HS102T	Universal Human Values	1	0	0	1	1	25	25	50	--	--	100
24HS103T	Indian Knowledge System	2	0	0	2	2	25	25	50	--	--	100
	Total	**14**	**1**	**10**	**25**	**20**						

24MA101T					Mathematics - I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To be able to evaluate problems related to differential and integral calculus of complex functions.
- To be able to obtain area and volume using integral calculus.
- To be able to formulate and solve various engineering problems using calculus.
- To study the properties of sequence and series and to check the convergence and divergence.

Unit I: DIFFERENTIAL CALCULUS AND ITS APPLICATIONS **Hours: 10**

Partial derivatives and their applications, Total derivatives, Jacobians, Maxima and minima of functions of two variables, Lagrange's method of undetermined multipliers, Power series and their convergence.

Unit II: INTEGRAL CALCULUS AND ITS APPLICATIONS **Hours: 15**

Definition and evaluation of double integrals (Cartesian and polar form), Change of order of integration, Change of variables, Evaluation of triple integrals, Change of variables (Cartesian to spherical and cylindrical), Applications to area, volume, centre of mass and centre of gravity by double and triple integrals.

Unit III: VECTOR CALCULUS **Hours: 12**

Gradient, divergence and curl, Directional derivative, Irrotational and solenoidal vector fields, Vector integration, Simple problems on line, surface and volume integrals, Green's theorem, Gauss divergence theorem and Stokes' theorem (without proofs), Simple applications involving cubes and rectangular parallelepipeds.

Unit IV: SEQUENCE AND SERIES **Hours: 15**

Definitions; convergent, divergent, bounded and monotone sequences; Infinite series: basics, Taylor series, convergence and divergence, tests for convergence for positive term series, alternating series, Leibnitz test, absolute convergence, conditional convergence.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the use of convergence of infinite series in engineering aspects.
CO2 - Understand the concept of directional derivative, irrotational and solenoidal vector fields.
CO3 - Apply the concepts of differential and integral calculus to engineering problems.
CO4 - Analyze the obtained solution in linear and non-linear domains.
CO5 - Appraise mathematical problems from the complex domain.
CO6 - Formulate and solve engineering problems using multivariable and vector calculus.

TEXT / REFERENCE BOOKS

1. Grewal, B. S. (2017). *Higher engineering mathematics* (44th ed.). Khanna Publishers.
2. Jain, R. K., & Iyengar, S. R. K. (2016). *Advanced engineering mathematics* (5th ed.). Narosa Publishing House.
3. Kreyszig, E. (2015). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
4. Strang, G. (2006). *Linear algebra and its applications* (4th ed.). Cengage Learning.
5. Hoffman, K., & Kunze, R. A. (2015). *Linear algebra* (2nd ed.). Prentice Hall of India.

24HS101T					English Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the fundamental elements of communication in the English language.
- To develop correct expression in the English language at a basic level.
- To use appropriate vocabulary, grammar and effective paragraph construction in day-to-day scenarios, including digital platforms.
- To learn and apply communication skills in different public and interpersonal contexts.

Unit I: INTRODUCTION TO COMMUNICATION **Hours: 6**

The fundamentals of language and communication; Why English?; Types of communication; Barriers to effective communication; Digital communication.

Unit II: ENGLISH GRAMMAR AND VOCABULARY **Hours: 6**

Tenses, articles, prepositions, active and passive voice, concord; Tools for vocabulary building; Homophones, homonyms, one-word substitution, antonyms, synonyms, root words, prefixes and suffixes, connotations, collocations, idioms and phrases.

Unit III: RECEPTIVE SKILLS **Hours: 7**

Listening skills: difference between listening and hearing, active and passive listening, types of listening, traits of a good listener. Reading skills: importance of reading, effective reading techniques, speed reading, the SQ3R method.

Unit IV: COMPENDIUM SKILLS **Hours: 7**

Note taking and note making: physical, digital and collective; Summarizing; Creating e-content.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate an understanding of the fundamentals of communication, including the basic components of the communication process and its significance in personal, academic and professional contexts.
CO2 - Apply grammatical rules accurately in written and spoken communication to enhance clarity, coherence and precision.
CO3 - Employ effective strategies for vocabulary development, such as word association, context clues and mnemonic devices.
CO4 - Identify and interpret implicit meanings, inference and figurative language in written and spoken texts.
CO5 - Develop written communication skills to convey complex ideas clearly, concisely and coherently.
CO6 - Apply ethical communication principles and standards of academic integrity while creating and sharing compendiums, avoiding plagiarism.

TEXT / REFERENCE BOOKS

1. Kumar, K. B. (2022). *Effective communication skills*. Khanna Book Publishing.
2. Wood, F. T. (2007). *Remedial English grammar*. Macmillan.
3. Sharma, S., & Mishra, B. (2009). *Communication skills for engineers and scientists*. PHI Learning.
4. Kaul, A. (2006). *Business communication*. Prentice-Hall of India.

24CP101T					Computer Programming - I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the usage of operators and data types.
- To apply different types of conditional and looping statements.
- To create different types of data collections and implement user-defined functions.
- To perform different operations upon files.

Unit I: BASICS OF PROGRAMMING **Hours: 3**

Introduction to computer programming, Features of C language, Structure of a C program, Program execution flow, C tokens, Variables, Data types, Operators, Decision control statements - if, switch, goto; Loop control structures - while, do-while, for; Break and continue statements.

Unit II: DERIVED DATA TYPES **Hours: 3**

Array: one-dimensional and multidimensional arrays, declaration, initialization, array manipulations, matrix operations; Strings - basic concepts, inbuilt string manipulation functions; Pointers, pointer arithmetic, pointer to pointer, array of pointers.

Unit III: FUNCTIONS AND STRUCTURES **Hours: 4**

Introduction to user-defined functions, types of functions, call by value and call by reference, recursion, pointers to functions, structures, array of structures, union.

Unit IV: FILE HANDLING **Hours: 3**

File handling in C, different types of files, operations on files such as file creation, deletion, access modes (read, write, append), file concatenation, file handling using the seek function.

Max 13 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of programming.
CO2 - Identify different programming constructs for a problem.
CO3 - Apply appropriate derived data types for data storage.
CO4 - Prepare a user-defined data type based on data attributes.
CO5 - Construct user-defined functions for problem solving.
CO6 - Analyse different data structures based on application requirements.

TEXT / REFERENCE BOOKS

1. Kernighan, B. W., & Ritchie, D. M. (2015). *The C programming language* (2nd ed.). Prentice Hall of India.
2. King, K. N. (2008). *C programming: A modern approach* (2nd ed.). W. W. Norton.
3. Griffiths, D., & Griffiths, D. (2012). *Head first C: A brain-friendly guide*. O'Reilly Media.
4. Balagurusamy, E. (2019). *Programming in ANSI C* (8th ed.). McGraw-Hill.
5. Kanetkar, Y. P. (2016). *Let us C* (15th ed.). BPB Publications.

24CP101P					Computer Programming - I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To implement basic programming concepts.
- To create different types of data collections.
- To implement user-defined functions.
- To perform different file handling operations.

LIST OF EXPERIMENTS

1. Introduction to computer programming: understanding the compilation process through a simple C program, program execution flow, C tokens, variables, keywords and identifiers, types of C constants and variables.
2. Simple and formatted input-output operations: data types, operators, input/output statements in C, formatted I/O, format specifiers, escape sequences.
3. Decision making and branching: if, if-else, if-else ladder, switch, goto statement, conditional operator statement.
4. Looping control structures: while, do-while, for loop, break statement, continue statement.
5. Derived data types - arrays and strings: one-dimensional and multidimensional arrays, declaration, initialization, array manipulations, matrix operations, string manipulation with and without inbuilt functions.
6. Derived data types - structure and union: structure, arrays and structures, structures and functions, pointer to structure, typedef, unions.
7. Functions: user-defined functions, types of functions, call by value and call by reference, header file creation, recursion, pointers to functions, arrays and functions.
8. Pointers: basics, use of & and * operators, pointer arithmetic, array and string processing using pointers, pointer to pointer, array of pointers.
9. File handling in C: different types of files, operations on files, file handling functions.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the use of appropriate naming conventions and programming style, including appropriate comment density.
CO2 - Implement a basic C program using appropriate control structures.
CO3 - Apply appropriate derived data types based on data attributes.
CO4 - Develop solutions with pointers and utilize them to access strings and structures.
CO5 - Design user-defined functions for problem solving and reuse them across different programs.
CO6 - Apply suitable file handling functions and operations.

TEXT / REFERENCE BOOKS

1. Kernighan, B. W., & Ritchie, D. M. (2015). *The C programming language* (2nd ed.). Prentice Hall of India.
2. King, K. N. (2008). *C programming: A modern approach* (2nd ed.). W. W. Norton.
3. Balagurusamy, E. (2019). *Programming in ANSI C* (8th ed.). McGraw-Hill.
4. Kanetkar, Y. P. (2016). *Let us C* (15th ed.). BPB Publications.

24CH101T					Engineering Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To develop a fundamental understanding of traditional materials.
- To provide knowledge about structural features, synthesis and properties of various categories of advanced materials.
- To develop knowledge and skills for different characterization techniques of materials.
- To provide knowledge about the role of chemistry in modern engineering applications.

Unit I: CHEMISTRY OF ENGINEERING MATERIALS **Hours: 12**

Traditional materials: introduction and classification of materials; metallic, polymeric and ceramic materials. Advanced materials: introduction to nanomaterials - properties and applications; carbonaceous materials (fullerene, carbon nanotube, graphene); composite materials; liquid crystals - classification and applications.

Unit II: MODERN ANALYTICAL TECHNIQUES **Hours: 9**

Instrumentation, principle and characterization of materials: X-ray diffraction (XRD); electroanalytical techniques; FTIR and UV-visible spectroscopy; thermal analysis (TGA-DTA-DSC); chromatographic techniques (GC, HPLC).

Unit III: ADSORPTION, CATALYSIS AND KINETICS **Hours: 9**

Adsorption - characteristics, classification and applications; adsorption isotherms - Freundlich, Langmuir and BET. Chemical kinetics - rate law, Arrhenius equation, transition state theory, collision theory, complex reactions. Catalysis - homogeneous and heterogeneous catalysis, mechanism of catalysis, industrial applications of catalysts.

Unit IV: CHEMISTRY OF ENERGY DEVICES **Hours: 9**

Principles of primary and secondary batteries; fuel cells and their operating principles; principles and uses of supercapacitors; photocatalytic hydrogen production - principles and challenges; traditional and new-generation solar cells.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the structural features and properties of different classes of traditional materials.
CO2 - Classify advanced materials such as nanomaterials, carbonaceous and composite materials and liquid crystals.
CO3 - Apply the understanding of various instrumental techniques for the characterization of materials.
CO4 - Analyze key concepts in engineering chemistry, viz. adsorption and chemical kinetics, and their applications to engineering challenges.
CO5 - Justify the industrial applications of different types of catalysis by analysing mechanisms of catalysis.
CO6 - Develop knowledge of the role of chemistry in modern engineering applications such as energy devices.

TEXT / REFERENCE BOOKS

1. Callister, W. D. (2007). *Materials science and engineering: An introduction* (7th ed.). John Wiley & Sons.
2. Barsoum, M. W. (2003). *Fundamentals of ceramics*. IOP Publishing.
3. Murty, B. S., Shankar, P., Raj, B., Rath, B. B., & Murday, J. (2013). *Textbook of nanoscience and nanotechnology*. Springer.
4. Raghavan, V. (2003). *Materials science and engineering: A first course* (4th ed.). Prentice-Hall of India.
5. Skoog, D. A., Holler, F. J., & Crouch, S. R. (2014). *Principles of instrumental analysis* (6th ed.). Cengage Learning.
6. Jain, P. C., & Jain, M. (2015). *Engineering chemistry* (16th ed.). Dhanpat Rai Publishing Company.

24CH101P					Engineering Chemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To enhance and develop scientific and analytical skills.
- To relate concepts learned in chemistry and engineering to real-world situations.
- To acquire skills to perform laboratory experiments.
- To demonstrate safe and proper use of standard chemistry glassware and equipment.

LIST OF EXPERIMENTS

1. Iodometry: to determine the strength of a given copper sulphate solution by titrating against N/20 sodium thiosulphate (hypo) solution.
2. Iodimetry: to determine the strength of given ascorbic acid by titrating against standard N/10 iodine solution.
3. Complexometric titration: to determine the total, permanent and temporary hardness of given water using standard 0.01 M EDTA solution.
4. pH-metric titration: to determine the strength of a given HCl solution using a standard NaOH solution.
5. Conductometric titration: to determine the strength of a given HCl solution using a standard NaOH solution.
6. Chemical kinetics: to study the kinetics of decomposition of sodium thiosulphate by a mineral acid.
7. Drawing chemical structures: to draw chemical structures of organic molecules using ChemDraw.
8. Colorimetric determination: to determine the concentration of copper present in the effluent of electroplating industries using a colorimeter.
9. Detection of biomolecules.
10. Preparation of a drug molecule: preparation of aspirin from salicylic acid.
11. Polymerization: to prepare a polymer (Nylon 6,10) and identify the functional groups by FT-IR.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the concepts learned in chemistry and engineering in real-world situations.
CO2 - Show the ability to identify, analyse and interpret the results of experiments.
CO3 - Experiment with instrumental methods using a conductometer and a pH meter.
CO4 - Analyse compounds by titrimetric, gravimetric and instrumental methods.
CO5 - Determine the concentration of unknown solutions by spectrophotometric methods.
CO6 - Predict the reaction rate, order and rate constant.

TEXT / REFERENCE BOOKS

1. Ahluwalia, V. K., Dhingra, S., & Gulati, A. (2005). *College practical chemistry*. Universities Press.
2. Baruah, J. B., & Gogoi, P. (2018). *Foundations of experimental chemistry*. PharmaMed Press.
3. Sawhney, S. S., Jassal, M. S., & Mittal, S. P. (2010). *A text book of chemistry practical* (Vols. 1-2). APH Publishing Corporation.

24EE101T					Elements of Electrical and Electronics Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart knowledge on DC and AC circuits.
- To learn the construction, working principle and characteristics of transformers and induction machines.
- To introduce students to various means for electrical safety, protection of electrical installations and batteries.
- To understand rectification through the p-n junction diode, applications of diodes and transistor characteristics.

Unit I: BASICS OF ELECTRONICS **Hours: 9**

Semiconductor diodes and applications, diode as a switch, diode as a rectifier, half-wave and full-wave rectifiers with and without filters; breakdown mechanisms, Zener diode operation and applications; opto-electronic devices - LEDs, photo diode and applications; silicon controlled rectifier (SCR) - operation, construction, characteristics, ratings and applications. Transistor characteristics: bipolar junction transistor (BJT) construction, operation, amplifying action, common base, common emitter and common collector configurations; field effect transistor (FET) construction and characteristics of junction FET.

Unit II: DC CIRCUITS **Hours: 9**

Electrical circuit elements (R, L and C), voltage and current sources, dependent and independent sources, Ohm's law, temperature coefficient of resistance, Kirchhoff's current and voltage laws, voltage and current divider circuits, Thevenin and Norton theorems and their equivalents, maximum power transfer and superposition theorems, nodal and mesh analysis, star-delta transformation, time domain analysis / natural response of first order RL and RC circuits.

Unit III: AC CIRCUITS **Hours: 9**

Generation of AC voltage, representation of sinusoidal waveforms, RMS values of different sinusoidal waveforms, rectangular and polar representation of phasors, sinusoid representation in time and frequency domain, analysis of single-phase AC series circuits consisting of R, L, C, RL, RC and RLC combinations, instantaneous power, average power and reactive power, complex power and power factor; AC parallel circuits and their solution in admittance form, resonance in AC series and parallel circuits, polyphase circuits, star and delta representation, power measurement in polyphase circuits.

Unit IV: INTRODUCTION TO ELECTRICAL MACHINES AND ELECTRICAL INSTALLATION **Hours: 12**

Magnetic materials and their B-H characteristics. Single-phase transformers: construction, working principle, types, EMF equation, ideal and practical transformer, operation on no-load and load. Induction machines: construction, types of 3-phase induction motors, working principle, production of rotating magnetic field, operation, starting and running torques, torque-slip characteristics, power stages. Components of LT switchgear: switch fuse unit (SFU), MCB, ELCB, MCCB; types of wires and cables; earthing; types of batteries and their important characteristics; elementary calculations for energy consumption; electrical safety rules.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate the application of different diodes in circuits and understand transistor characteristics.
CO2 - Understand and analyse electrical circuits using network theorems.
CO3 - Compare the behaviour of R, L and C and their combinations in AC circuits.
CO4 - Analyze balanced polyphase systems in star and delta configurations.
CO5 - Understand the construction, working and basic characteristics of transformers and induction machines.
CO6 - Recognize the importance of protective devices, electrical safety measures and characteristics of batteries.

TEXT / REFERENCE BOOKS

1. Bird, J. (2017). *Electrical circuit theory and technology* (6th ed.). Routledge, Taylor and Francis Group.
2. Kothari, D. P., & Nagrath, I. J. (2019). *Basic electrical engineering* (4th ed.). Tata McGraw Hill.
3. Theraja, B. L., & Theraja, A. K. (2014). *A textbook of electrical technology* (Vol. 1). S. Chand Publication.
4. Singh, S. (2018). *Electrical engineering materials*. Dhanpat Rai and Co.
5. Boylestad, R. L., & Nashelsky, L. (2015). *Electronic devices and circuit theory* (11th ed.). Pearson Education.

24EE101P					Elements of Electrical and Electronics Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To gain practical knowledge of DC and AC circuits.
- To learn the operation of electrical instruments and electrical machines.
- To introduce students to various means for electrical safety and protection of electrical installations.
- To understand characteristics and applications of semiconductor diodes and transistors.

LIST OF EXPERIMENTS

1. Introduction to the elements of electrical engineering laboratory and study of different electrical measuring instruments.
2. To verify Kirchhoff's current and voltage laws for DC circuits.
3. To validate Thevenin and Norton theorems for DC circuits.
4. To validate superposition and maximum power transfer theorems for DC circuits.
5. To evaluate the AC R, L and R-L series circuit performance and to measure active, reactive and apparent power with single-phase AC supply.
6. To evaluate the performance of an AC R-L parallel circuit and to measure active, reactive and apparent power with single-phase AC supply.
7. To perform a direct load test on a single-phase transformer and determine efficiency and voltage regulation at different loads.
8. To measure three-phase power using the two-wattmeter method.
9. To draw the time-current characteristics of an MCB and study the function of an ELCB.
10. To obtain the V-I characteristics of silicon and germanium diodes.
11. To obtain the reverse characteristics of a Zener diode.
12. To study half-wave, full-wave and bridge rectifiers.
13. To determine the DC characteristics of a BJT in CE configuration.
14. To study the types of batteries and their characteristics.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify and operate different electrical measuring instruments.
CO2 - Perform and verify various network theorems for DC circuits.
CO3 - Understand the performance of AC circuits with R and L loads.
CO4 - Evaluate star and delta configurations of polyphase systems and measure power in polyphase systems.
CO5 - Evaluate the performance of a single-phase transformer.
CO6 - Understand the working principle of semiconductor diodes, transistor characteristics and their applications.

TEXT / REFERENCE BOOKS

1. Bird, J. (2017). *Electrical circuit theory and technology* (6th ed.). Routledge, Taylor and Francis Group.
2. Kothari, D. P., & Nagrath, I. J. (2019). *Basic electrical engineering* (4th ed.). Tata McGraw Hill.
3. Boylestad, R. L., & Nashelsky, L. (2015). *Electronic devices and circuit theory* (11th ed.). Pearson Education.

24BT101T					Biological Systems for Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the importance of bioengineering principles, advanced biomedical technologies and emerging trends in the field.
- To introduce students to biodesign principles for creating novel devices and structures.
- To develop the skills necessary for designing innovative solutions inspired by biological systems.
- To evaluate the diagnostic capabilities and limitations of various imaging modalities and molecular diagnostic techniques.

Unit I: INTRODUCTION TO BIOMOLECULES **Hours: 6**

Cell as the basic unit of life; overview of cell structure and functions. Carbohydrates: properties and biological functions. Nucleic acids: structure and roles of DNA and RNA. Proteins: structure and functions. Significance of lipids. Role of biomolecules: characteristics and roles of enzymes, vitamins and hormones.

Unit II: INSPIRING BIOENGINEERING DESIGN **Hours: 6**

Functions and classification of the nervous system; the brain as a central processing unit - concepts of artificial neural networks and machine learning techniques; the eye as a camera; the heart as a circulatory system; the ear and nose as signal transmission models; electronic nose, electronic tongue, electronic skin; lungs as an exchange model and kidneys as a filtration system.

Unit III: ADVANCED IMAGING AND DIAGNOSTICS **Hours: 7**

Introduction to radiology - X-ray, ultrasound, EEG, CT, PET; high-throughput diagnostics in clinics; molecular diagnostics - PCR, DNA chip, diagnosis of genetic disorders.

Unit IV: TRENDS IN BIOENGINEERING **Hours: 7**

Stem cells and their applications; bioprinting techniques and materials; applications of bioinformatics; artificial intelligence for diagnosis and detection of communicable and non-communicable diseases; biosensors in healthcare; environmental monitoring.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental principles of cellular biology, including the structure and functions of cells, genes and biomolecules.
CO2 - Demonstrate the involvement of major organs and systems of the human body as models for engineering design.
CO3 - Analyse the scope and opportunities of implementing bioengineering concepts in advanced imaging and diagnostic techniques.
CO4 - Get acquainted with concepts of bioengineering trends including biosensors, AI and imaging techniques in disease diagnosis and treatment.
CO5 - Focus on the implications of bio-inspired designs, tissue engineering and molecular diagnostics.
CO6 - Design and develop innovative solutions for healthcare, environmental monitoring and biosecurity.

TEXT / REFERENCE BOOKS

1. Thyagarajan, S., Selvamurugan, N., Rajesh, M. P., Nazeer, R. A., Thilagaraj, W., Barathi, S., & Jaganthan, M. (2012). *Biology for engineers*. Tata McGraw-Hill.
2. Johnson, A. T. (2011). *Biology for engineers*. CRC Press, Taylor and Francis.
3. Cooper, G. M. (2016). *The cell: A molecular approach* (7th ed.). ASM Press.
4. Singh, S., & Allen, T. (2014). *Biology for engineers*. Vayu Education of India.
5. Bar-Cohen, Y. (2012). *Biomimetics: Nature-based innovation* (1st ed.). CRC Press.
6. Fox, S., & Rompolski, K. (2022). *Human physiology* (16th ed.). McGraw-Hill.
7. Floreano, D., & Mattiussi, C. (2008). *Bio-inspired artificial intelligence: Theories, methods and technologies*. MIT Press.

24HS104T					Organizational Behaviour					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

- To enable the development of a perspective to diagnose and efficiently deal with matters of human behaviour in organizations.
- To enrich understanding of the dynamics of interaction and integration between the individual and the organization.
- To explore the organization system in entirety to capture the full complexity of organizational behaviour.
- To critically appraise the potential effects of important developments in the external environment on organizational behaviour.

Unit I: INTRODUCTION TO ORGANIZATIONAL BEHAVIOUR **Hours: 3**

Definition and scope of organization, behaviour and organizational behaviour; historical developments and emerging concerns in OB; perception in organizational behaviour - elements, principles and biases; learning theories, conditioning and application in organizations; personality theories; attitudes in the workplace.

Unit II: EMOTIONS, STRESS MANAGEMENT, GROUP DYNAMICS AND INTERPERSONAL RELATIONSHIPS **Hours: 4**

Understanding emotions in organizational behaviour; emotional labour; stress management techniques in organizational settings; group formation and dynamics; group decision-making techniques and their application; team building strategies and effective communication; influence, power dynamics and organizational politics.

Unit III: LEADERSHIP AND ORGANIZATIONAL CULTURE **Hours: 3**

Leadership theories - overview and application in organizational settings; leadership styles and their impact on organizational behaviour; understanding organizational culture and climate.

Unit IV: ORGANIZATIONAL CHANGE AND STRUCTURE, BALANCING WORK AND LIFE **Hours: 3**

Organizational change processes; resistance management during organizational change; organizational structure and design; job satisfaction - determinants and impact on employee behaviour; stress prevention and management techniques; work-life balance strategies and their role in organizational behaviour.

Max 13 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Critically gauge the potential effects of important developments in the external environment on organizational behaviour.
CO2 - Analyze organizational behavioural issues in the context of organizational behaviour theories, models and concepts.
CO3 - Analyze the behaviour of individuals and groups in organizations in terms of the crucial factors that affect organizational behaviour.
CO4 - Examine organizational behavioural matters in the context of established theories and models.
CO5 - Demonstrate how organizational behaviour helps in understanding the motivation behind the behaviour of people in the organization.
CO6 - Manage conflict in the organizational context and deal with stress.

TEXT / REFERENCE BOOKS

1. Wilson, F. M. (2018). *Organizational behaviour and work* (5th ed.). Oxford University Press.
2. Robbins, S. P., Judge, T. A., & Vohra, N. (2019). *Organizational behaviour* (18th ed.). Pearson Education.
3. Aswathappa, K. (2019). *Organizational behaviour* (12th ed.). Himalaya Publishing House.
4. Luthans, F. (2011). *Organizational behavior: An evidence-based approach* (12th ed.). McGraw-Hill.

24YOG101					Yog, Health & Hygiene					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To impart to students the basic concepts of physical education, sports and yoga for health and wellness.
- To familiarize students with health-related exercises, sports and yoga for overall growth and development.
- To create a foundation for professionals in physical education, sports and yoga.
- To impart the basic knowledge and skills to teach physical education, sports and yoga activities.

LIST OF EXPERIMENTS

1. Kinesiology and cardio fitness test: introduction to kinesiology and the physiological basis of conditioning, sports psychology and the Cooper cardio fitness test (12-minute run/walk); expert tips, tools and training for walking and running.
2. Yoga: introduction to yogasana and yoga therapy as a rehabilitation tool; effect of yoga on exercise endurance as assessed by cardiorespiratory efficiency tests; yogic practices that promote and improve respiratory and cardiovascular function and enhance physical fitness.
3. Games and sports league: practice sessions for outdoor and indoor games, event-wise and game-wise practice as per the student's interest - football, chess, cricket, tennis, basketball, volleyball, athletics (relay) and pickleball.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Discover the balance of health and happiness through the basic principles and practices of physical education, sports and yoga.
CO2 - Understand physical activities, sports and yoga practices, theories and rules of various games for healthy living.
CO3 - Analyze emerging trends and issues in world sports and develop leadership qualities to conduct, organize and officiate physical education, sports and yoga events.
CO4 - Practise on the field and in the indoor yoga hall.
CO5 - Develop a spirit of teamwork and fair play.
CO6 - Demonstrate understanding by participating in games and sports leagues.

TEXT / REFERENCE BOOKS

1. Agola, R. (2018). *Athletic track and court marking handbook of games and sports*. Sports Publication.
2. Saraswati, S. S. (2013). *Asana, pranayama, mudra, bandha*. Yoga Publications Trust, Munger.
3. Pathak, P. V. (2016). *Sports games and rules, regulations*. Sports Publication.
4. Gharote, M. L. (2015). *Yogic prakriyanche margdarshan*. The Lonavala Yoga Research Institute.

24NSS101					National Service Scheme (NSS)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To develop a sense of civic and social responsibility.
- To identify the needs and problems of the community and involve them in problem solving.
- To engage in creative and constructive social action.
- To develop social character and leadership through NSS.

LIST OF EXPERIMENTS

1. Environment and sustainability - water and waste management: volunteering work for environment and sustainability (water and waste management) and tree plantation.
2. NSS seven-day special camp: volunteering for tree planting, agricultural compost, tree guards, Gujarat Skill Development Mission and social activities in the village as per the government NSS manual.
3. Fit India Mission: volunteering for cardio fitness, yoga, running, Mission Olympics, self-defence and Agneepath Mission.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the needs and problems of the community.
CO2 - Understand the importance of his/her responsibilities towards society.
CO3 - Analyze environmental and societal problems and issues.
CO4 - Evaluate the existing system and propose practical solutions for sustainable development.
CO5 - Develop government or self-driven projects effectively in the field.
CO6 - Understand the implementation of government or self-driven projects in society.

TEXT / REFERENCE BOOKS

1. National Service Scheme Unit, PDEU. (2023). *NSS course manual*. Pandit Deendayal Energy University.
2. Government of Gujarat. (2023). *NSS cell activity guidelines*. NSS Cell, Government of Gujarat.
3. Government of India. (2023). *NSS manual and activity reports*. Ministry of Youth Affairs and Sports.

24NCC101					National Cadet Corps (NCC)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To develop discipline, character, brotherhood, the spirit of adventure and ideals of selfless service amongst young students.
- To develop youth leadership qualities in the students.
- To induce social consciousness among students through various NCC camps.
- To inculcate the foundations of drill, weapon training and adventure activities.

LIST OF EXPERIMENTS

1. Introduction to NCC: aims and objectives, structure and organization of NCC, cardinals of NCC, NCC flag, oath of NCC, NCC song, incentives of NCC.
2. National integration: importance of national integration and awareness, necessity, national interests, objectives, threats and opportunities, unity in diversity.
3. NCC camps, social service, awareness and community development activities: social awareness and community development, health and hygiene, environment awareness and conservation; participation in blood donation camps, Swachhata Abhiyan, Constitution Day; participation in NCC camps such as ATC, CATC, NIC, COC, TSC, RDC and leadership camps.
4. Drill, weapon training and adventure activities: types of drill, foot drill, general and words of command, saluting, weapon training, map reading, field craft and battle craft, introduction to infantry weapons and equipment, obstacle and weapon training, adventure training, participation in Republic Day and Independence Day ceremonial parades at the university.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Know the history of NCC, its organization and the incentives of NCC for career prospects.
CO2 - Understand the importance of nation building and individual contribution to the same.
CO3 - Maintain discipline and team spirit.
CO4 - Build character and leadership qualities.
CO5 - Understand drill as the foundation for discipline and command a group towards a common goal.
CO6 - Develop a sense of selfless social service for better social and community life.

TEXT / REFERENCE BOOKS

1. Directorate General of NCC. (2020). *Cadet handbook SD/SW - Common subjects, all wings*. DG NCC, New Delhi.